

Short Report on Dataset Preprocessing Steps

1. Dataset Information Extraction

The first preprocessing step involved analyzing the raw CSV dataset to understand its structure and quality. The dataset was loaded using the Pandas library, and basic statistics were generated, including total number of rows, columns, null values, and class distribution of the output label column.

The output column (last column) was specifically analyzed to identify:

- Number of unique output classes
- Count of each class
- Presence of null values

For numeric columns, statistical measures such as:

- Minimum value
- Maximum value
- Mean value

were computed. Additionally, value distribution was examined using bin-based range analysis to understand feature spread.

This step helped identify:

- Missing values
- Infinite values
- Class imbalance
- Data consistency issues

2. Dataset Cleaning and Null Value Handling

The dataset contained missing and infinite values, particularly in the **Flow Bytes/s** column.

Preprocessing actions performed:

- Replaced infinite values ($+\infty$, $-\infty$) with NaN
- Segregated rows with null and non-null values
- Used **XGBoost Regressor** to predict missing values in the Flow Bytes/s column
- Filled remaining missing values using **median imputation**

This ensured that the dataset became complete and suitable for machine learning.

3. Bias Reduction and Dataset Balancing

The dataset showed class imbalance between **BENIGN** and attack traffic classes.

To reduce model bias:

- Applied **SMOTE (Synthetic Minority Over-sampling Technique)**

SMOTE generated synthetic samples for the minority class by creating slight variations from existing samples rather than exact duplication.

This resulted in:

- Balanced class distribution
- Reduced prediction bias
- Improved training reliability

4. Final Output

After preprocessing, the final dataset had:

- No null values
- No infinite values
- Balanced class distribution
- Clean numerical consistency

The processed dataset was saved as a new CSV file and prepared for machine learning model training.

Conclusion

The preprocessing phase improved dataset quality by extracting structural information, cleaning missing/infinite values, and reducing class imbalance. These steps enhanced data reliability and prepared the dataset for accurate intrusion detection model development.